Research Methods — Module 05: Action — Study Questions

1. How does Active Inference replace classical inverse kinematics for robot control?
2. What are the advantages of a unified perception-action framework in robotics?
3. Describe pymdp's core data structures. What are A, B, C, D matrices?
4. How does RxInfer.jl differ from pymdp? What applications does each serve best?
5. What are generalized coordinates of motion? Why are they needed for continuous control?
6. How is free energy computed in generalized coordinates?
7. What are attractor dynamics? How do they encode desired behavior?
8. How does a robot balance exploration and exploitation during navigation?
9. How does precision weighting determine the balance between visual and tactile feedback?
10. How can a robot model a human partner as another Active Inference agent?
11. What is the implementation workflow from model definition to hardware deployment?
12. What challenges arise when transferring from simulation to real hardware?
13. How does Active Inference handle unexpected perturbations during robot manipulation?
14. Compare Active Inference control with PID control for a reaching task.
15. Compare Active Inference control with model-based RL for navigation.
16. What is deep active inference? How do neural networks approximate the generative model?
17. How do you validate that a robot is "truly" performing Active Inference?
18. What precision parameters must be tuned for stable robot behavior?
19. How does Active Inference handle sensor noise and actuator uncertainty?
20. Design a robot experiment that demonstrates Active Inference's advantage over classical control.